

ME350 FA 2025 Section 002

Lab #6

Monday 12:30 PM - 2:30 PM

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Office Hours: 3:00 PM - 5:00 PM Monday (Zoom)



Agenda

- Reminders
- Project Schedule
- Gate Review 2
- Lab7 Functional Testing
- Mid Semester Peer Evaluation



Reminders and Project Schedule

Oct 5	Oct 6	Oct 7	Oct 8	Oct 9	Oct 10
		Lecture 12 - Transmission Gear Forces (MU)		Lecture 13 - Transmissions: Gears - Strength and Wear (MU)	
	Lab 6 - Gate Review 2 in Lab (X50 Assembly Room, 1510 GGB) Lab Assignment #5 Due - Transmission Selection Workbook Bill of Materials (BOM) Transmission CAD			Lab 7 - Functional Testing (X50 Assembly Room, 1510 GGB) Lab Assignment #6 Due - Manufactured GR1 Parts	Lab 7 - Functional Testing (X50 Assembly Room, 1510 GGB) Lab Assignment #6 Due - Manufactured GR1 Parts
<-- Build & Assembly for Functional Testing					
Oct 12	Oct 13	Oct 14	Oct 15	Oct 16	Oct 17
		No Lecture -- Fall Study Break		Lecture 14 - Intro to Mechatronics (JK)	
	No Lab -- Fall Study Break Lab 7 - Functional Testing TO BE COMPLETED ASYNCHRONOUSLY FOR MONDAY LABS Lab Assignment #6 Due - Manufactured GR1 Parts			- Homework 4 Due (Transmissions 1) - Homework 5 Released Lab 8 - Sensors and Microcontrollers 1: Wiring, switches, and IR Prox sensor Lab Assignment #7 Due - Manufactured GR2 Parts	Lab 8 - Sensors and Microcontrollers 1: Wiring, switches, and IR Prox sensor Lab Assignment #7 Due - Manufactured GR2 Parts

Gate Review 2

Goal: Obtain GSI approval to proceed with transmission design and manufacturing / ordering your BOM.

- Quick presentation and CAD review
- Location: In Lab 6
- 15-20 minute time slot
- Remember to have your CAD ready!

Time Slot	Presenting Team
12.45 pm - 1.00 pm	Team #1
1.00 pm - 1.15 pm	Team #2
1.15 pm - 1.30 pm	Team #3
1.30 pm - 1.45 pm	Team #4
1.45 pm - 2.00 pm	Team #5
2.00 pm - 2.15 pm	Team #6
2.15 pm - 2.30 pm	Team #7



Parts for Assembling your Linkage

- Thrust washers are in the gray cabinet
- Shoulder bolts are in the gray cabinet
 - 10-32 shoulder bolts exist but are available in less quantities than the 10-24 bolts
- Steel washers are available in multiple locations
 - Some in X50 in the bins
 - Majority of washers available in machine shop
- Regular 10-24 bolts available in machine shop
- Lots of hardware available in the machine shop
- New large tool chest holds all tools in the ME Assembly room
 - Please return all tools to this chest!



Limit Switch Mounting

- Consider how you are going to attach the limit switch(s) to your linkage/baseplate.
 - Limit switch is needed for Lab 9 to calibrate linkage encoder count (linkage position)
 - It must contact a link (input or follower) at one end of the full range of motion. You can use two limit switches (one for each limit) to improve encoder accuracy.
 - The switch needs to compress when your linkage hits a hardstop (consider adjustability)
 - The [CAD is available here](#)



Before Lab 7, in order of priority:

- Assemble linkage
 - If you have not added the holes for your input link because you are waiting to finalize your input shaft size, no problem!
- Follow up on your GR1 drawings/MPs that haven't been approved
- Work on GR2 Parts
 - If you're resizing input link holes in your drawing: write in the new hole size for your input shaft, ask Mike Klein/Don/Jon/Kemal to initial the update
 - If you're adding a new hole for a spring-pin, dowel-pin, or set-screw: create a new drawing
 - Start drawings/MPs for transmission parts



Mid Semester Peer Evaluation

- Mid Semester Peer Evaluation Guideline: [Link](#)
- Each student receives 100 contribution points and will distribute these points to themselves and their teammates
- Evaluation should be based on
 - Professionalism
 - Work Quantity
 - Work Quality
- Project grade will be adjusted based on evaluation score

Contribution points	Effect on final grade
< 60	-05
60-66	-04
66-72	-03
72-80	-02
80-92	-01
92-108	0
108-120	+01
120-128	+02
128-134	+03
134-140	+04
>140	+05

X50 Assembly Room 3D Printing

- Locally download the Bambu Lab Slicer, which is compatible with the X50 Assembly Room printers
 - Can be found here: <https://bambulab.com/en/download/studio>
- Download your part as a .STL File and upload to Bambu Lab
- Slice your part
- To obtain approval for your sliced file, either:
 - Attend office hours to review file and obtain official approval from a **GSI/IA**
 - Visit Kemal's office (1531A GGBL) and obtain official approval from Kemal
- Visit Kemal's Office (1531A GGBL) to check out a key to a printer anytime M-F from 8:30 AM - 4:30 PM. You will trade your MCard for the Key.
- Print on printers in X50 Assembly Room
- Return Key immediately after print completion or risk losing access to X50 Assembly room printers



Today

- Present Gate Review 2

Time Slot	Presenting Team
12.45 pm - 1.00 pm	Team #1
1.00 pm - 1.15 pm	Team #2
1.15 pm - 1.30 pm	Team #3
1.30 pm - 1.45 pm	Team #4
1.45 pm - 2.00 pm	Team #5
2.00 pm - 2.15 pm	Team #6
2.15 pm - 2.30 pm	Team #7

Due next lab:

- Manufacture all parts from Gate Review 1
- Assemble linkages for testing

